

Ayoub Alouan

Software Engineer · Java, Python, AI & Quality

Java / Spring Boot · Python / Flask · LLMs & AI agents · React · Test automation · Docker / AWS

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Summary

State-certified software engineer (EMSI, 2025) with a versatile profile: **Java / Spring Boot** back end, data and AI in **Python, React / Angular** front ends and **software quality**. Nearly one year on a production data platform: **microservices** (1,000 req/min), **ETL pipelines** (500K+ leads/day), a **multi-agent LLM** application, **unit, integration and API tests** and **Docker / AWS** deployment.

Technical skills

Java	Java SE (17/21), Spring Boot, Spring Cloud, Spring Security, JPA / Hibernate, Microservices, Keycloak, JWT, Maven
Python & AI	Flask, FastAPI, Django REST, Celery / Redis, Pandas, LLM APIs (OpenAI, Gemini), CrewAI, PyTorch, Hugging Face, OCR
Frontend	React, Next.js, Angular (17+), TypeScript, Tailwind CSS
Testing & quality	JUnit 5, Mockito, PyTest, Selenium, Postman, SonarQube; unit, integration, API and regression testing
Databases	PostgreSQL, MongoDB, Elasticsearch, MySQL, Oracle SQL, SQL Server
DevOps & methods	Docker, Jenkins (CI/CD), AWS (EC2, S3), Linux, Git / GitHub, Agile / Scrum

Experience

IAWEB.DEV – Havet Digital

Marrakech

Software Engineer · Full-time

Aug 2025 – Dec 2025

- **Multi-agent** lead-generation application driven by **natural-language queries** (CrewAI, ChatGPT API, Flask, Next.js).
- Took **DataPull** to production: stabilisation, **Docker**, deployment on **AWS (EC2, S3)**.
- **Unit and integration tests** (JUnit 5, Mockito) and **API tests** (Postman), code quality with **SonarQube**.

IAWEB.DEV – Havet Digital

Marrakech

Full-Stack Engineer · Final-year internship

Mar 2025 – Aug 2025

DataPull – data- and AI-driven B2B/B2C lead-generation platform

- **Spring Boot** back end as **microservices** (Gateway, B2B, B2C, Users) handling up to **1,000 req/min**, **Keycloak** security (SSO, RBAC, JWT).
- **Python ETL pipelines** (Flask, Celery / Redis): cleaning and enriching **500K+ leads/day**.
- **Real-time** search with **Elasticsearch + PostgreSQL** and **React** front end (dashboards, map, exports).
- Unit, integration, API and regression **testing** (JUnit, Mockito, PyTest) before each release.

Court of Appeal of Marrakech · Web Developer · Internship

Jul 2024 – Sep 2024

- Leave management for **200+ employees** (Spring Boot, Angular, SQL Server): requests, approvals, notifications.

EKBlocks · Web Developer · Internship

Jul 2023 – Sep 2023

- School-management application in **Django REST** and **React**: enrolment, classes, teachers, grades and absences.

Projects

- **PatternHunter** (2026) – Design-pattern detection in Java code: comparison of **TF-IDF + SVM** (scikit-learn), **CodeBERT** and **LoRA** (PyTorch, Hugging Face). github.com/ALOUAN01/PatternHunter
- **AI Design Pattern Detector** – **Django, Spring Boot, Angular**: analyses a GitHub repository with the **Gemini** API and recommends design patterns.
- **ScannerCard** – **Django, Spring Boot, Angular**: extracts data from ID cards with **OpenCV**, **Tesseract** and **Google Vision**.

Education & languages

State Engineering Degree (Master's level) – Computer Science & Networks

EMSI Marrakech · 2020 – 2025

Certifications: Java Microservices (Spring Boot & Spring Cloud) · Machine Learning (Python) · Git / GitHub · DevOps

Languages: Arabic (native) · French (fluent) · English (advanced)